

MUHAMMAD ARHAM KHAN

SOFTWARE ENGINEER

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Innovative Full Stack Developer with expertise in Python, web development and artificial intelligence. Proven ability to build scalable solutions and automate tasks, with a strong foundation in backend programming and database management. Passionate about utilizing technology to solve complex problems.

TECH STACK

- Languages: Python, JavaScript, TypeScript
- Backend: Django, FastAPI, Flask, Node.js
- Databases: PostgreSQL, MongoDB, MySQL
- DevOps: Docker, NGINX, Heroku, Netlify
- Messaging: RabbitMQ, WebSockets
- AI/ML: OpenAI, NLP, ML models

EDUCATION

Bachelor of Science in Computer Science From Capital University of Science and Technology **2020 – 2024**

- Specialized in web development, AI, and cybersecurity.
- Developed a final year project focused on sentiment analysis using machine learning and NLP techniques.
- Coursework included: Data Structures, Algorithms, Artificial Intelligence, Database Systems, Software Engineering, and Cybersecurity.

PROFESSIONAL EXPERIENCE

Associate Software Engineer, Camden Health System (Pvt) Ltd **March 2025 - Present**

- Built and optimized Django APIs serving 100+ hospitals, improving response time by 30%
- Participating in code reviews, debugging, and optimizing applications for maximum performance.
- Working on API development, integration, and ensuring secure authentication and data exchange.
- Assisting in database design, management, and optimization for application needs.
- Supporting existing projects by identifying and fixing software issues.
- Built a full-stack video calling platform with WebRTC, Socket.IO, and React, featuring peer-to-peer calls, responsive UI, and real-time user display.
- Developed a scalable, Zoho/Gmail-like email platform with Django, PostgreSQL, and MongoDB, supporting individual/organization accounts, secure auth, and external mail server integration.
- Completed and enhanced a Lovable-built app by integrating Supabase backend, fixing workflows, and adding custom features for improved functionality.

Python Developer, GoldBarTech Pvt Ltd **July 2024 - March 2025**

- Developed an Amazon auto web scraper to automate product data collection every 12 hours, increasing data retrieval efficiency.
- Created an AI model for big data classification, improving data processing and analysis.
- Designed an SMTP email tool, streamlining automated bulk email communication.

Intern, Eziline Software House **Jul 2023 – Sep 2023**

- Gained hands-on experience in full-stack development using Angular and Nuxt.js frameworks.
- Assisted in building client websites, improving front-end designs and functionality.
- Collaborated with senior developers to improve backend systems and optimize database queries.
- Conducted testing and debugging of web applications to ensure functionality and user satisfaction.

Web Developer, Arctic Solution Ltd**Mar 2023 – Apr 2023**

- Led development of an advanced automation system, achieving a 15% increase in operational efficiency.
- Developed dynamic websites for small businesses, enhancing online presence and user experience.
- Designed and developed dynamic websites, boosting user engagement through responsive and interactive features.

Teacher, Skills Science College**Jul 2022 – Mar 2023**

- Taught computer science courses to high school students, focusing on web development and programming.
- Managed the computer science department's budget, gaining financial independence and responsibility.
- Improved classroom engagement through interactive teaching methods and real-world problem-solving exercises.
- Organized workshops and seminars to introduce students to emerging technologies like AI and machine learning.

Web Developer, Mak Traders**Jul 2021 – Nov 2021**

- Developed and maintained dynamic websites for multiple small businesses, improving online visibility.
- Provided technical support and troubleshooting for clients, ensuring optimal website performance.
- Integrated e-commerce solutions for small businesses, enabling online transactions and customer management.

PROJECTS & KEY ACHIEVEMENTS

- Built an AI-powered email management system using OpenAI and custom models to identify, classify, and manage emails for large-scale marketing businesses.
- Developed a PMTA server management agent (Flask) with DKIM/SPF generation, domain hosting, file management, server automation, and centralized control of distributed servers.
- Created Chrome extensions for Amazon USA/UK integrated with tools like Helium 10, enabling one-click scraping of product data (keywords, sellers, revenues, reviews, search volume) into Google Sheets for research and tracking.
- Built an automated Amazon product scraper running every 12 hours, improving data collection efficiency.
- Developed a Django-based conversational AI platform ("Sages") with human-like text/voice responses.
- Created a full-stack video calling platform (WebRTC, React, Socket.IO) supporting unlimited meetings, screen sharing, messaging, and responsive UI; also delivered a Dockerized therapy platform for patients to connect with doctors.
- Built a medical store management system with inventory, billing, and role-based access control; added monthly sales, analytics, and reporting.
- Designed a Python-based CCTV recognition app for real-time identity detection from video feeds.
- Scraped McDonald's USA branch locations with Selenium for business analytics.
- Developed a scalable, Gmail/Zoho-like email platform (Django, PostgreSQL, MongoDB) with multi-tenant architecture, secure auth, role-based access, and external mail server integration.
- Designed and deployed dynamic websites and Supabase-backed apps via Netlify, GitHub, and Heroku, fixing workflows and improving performance.
- Delivered a secure SMTP email-sending tool for automated communications.
- Successfully led large-scale university tech events, demonstrating organizational and leadership skills.